Lab Exercise 4- Building a Docker Image for an HTML App Using Nginx

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2. Step 1: Create the HTML File

Create a directory for your HTML app and place an index.html file in it.

```
mkdir nginx-html-app cd

nginx-html-app
```

Inside the nginx-html-app directory, create the HTML file.

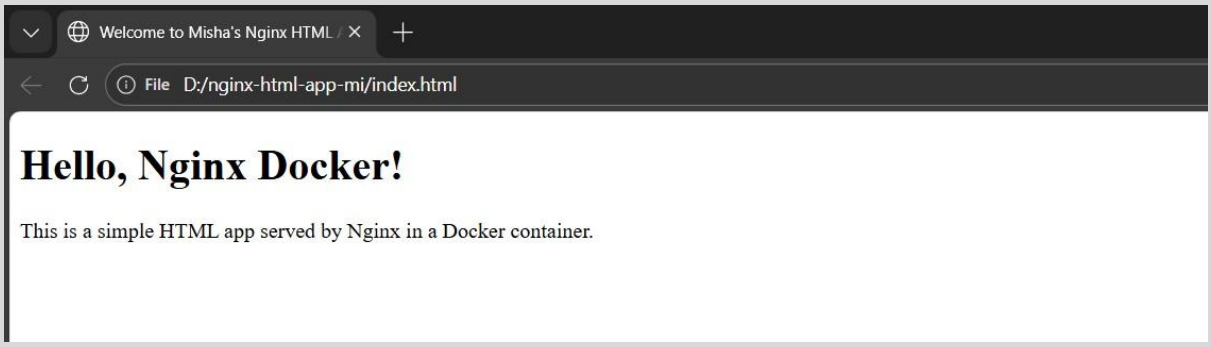
```
touch index.html
```

```
    + FullyQualifiedErrorId : CommandNotFoundException

PS D:\nginx-html-app-mi> New-Item index.html


    Directory: D:\nginx-html-app-mi


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        03-02-2026     11:20              0 index.html


PS D:\nginx-html-app-mi>
```

Edit the index.html file with the following content (or any custom HTML content you want):

```
<!DOCTYPE html>
<html>
<head>
   <title>Welcome to My Nginx HTML App</title>
</head>
```

```
<body>
  <h1>Hello, Nginx Docker!</h1>
  <p>This is a simple HTML app served by Nginx in a Docker container.</p>
</body>
</html>
```



3. Step 2: Create a Dockerfile

In the same directory, create a Dockerfile. This file will define how to build the Docker image using Nginx as the base image.

```
touch Dockerfile
```

Edit the Dockerfile and add the following content:

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/
EXPOSE 80
```

4. Step 3: Build the Docker Image

Now that you have the Dockerfile and index.html, it’s time to build the Docker image.

Run the following command to build the image, giving it a tag (e.g., nginx-html-app):

```
docker build -t nginx-html-app .
```

```
EXPOSE 80
PS D:\nginx-html-app-mi> docker build -t nginx-html-app-mi .
[+] Building 2.0s (7/7) FINISHED                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                  0.1s
 => => transferring dockerfile: 107B                                                  0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                       0.5s
 => [internal] load .dockerignore                                                     0.1s
 => => transferring context: 2B                                                       0.0s
 => [internal] load build context                                                     0.1s
 => => transferring context: 273B                                                     0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf  0.4s
 => => resolve docker.io/library/nginx:latest@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf  0.1s
 => [2/2] COPY index.html /usr/share/nginx/html/                                      0.1s
 => exporting to image                                                                0.5s
 => => exporting layers                                                               0.2s
 => => exporting manifest sha256:7a5126cc8d5febdba6d2130d50b15de46ac32c19157c42238ed1d82954fac298  0.0s
 => => exporting config sha256:505c0ba6bcb82e6a8794d6b8eef2752bb78caa5f717e5037d6383343a5a5c05e  0.0s
 => => exporting attestation manifest sha256:b1b336ec44ececab20967ca94928f58cce88507d1ad9cfd9f218d0e9bfee5ffc  0.1s
 => => exporting manifest list sha256:b184d0d81f70f644a89b4c40bbe9efc3d34ddc3161e45b19629bfb1f2eec3eac  0.0s
 => => naming to docker.io/library/nginx-html-app-mi:latest                           0.0s
 => => unpacking to docker.io/library/nginx-html-app-mi:latest                        0.1s
PS D:\nginx-html-app-mi>
```

Docker will use the Nginx base image, copy your index.html into the appropriate directory, and build the image.

5. Step 4: Run the Docker Container

After building the image, you can run the container with the following command:

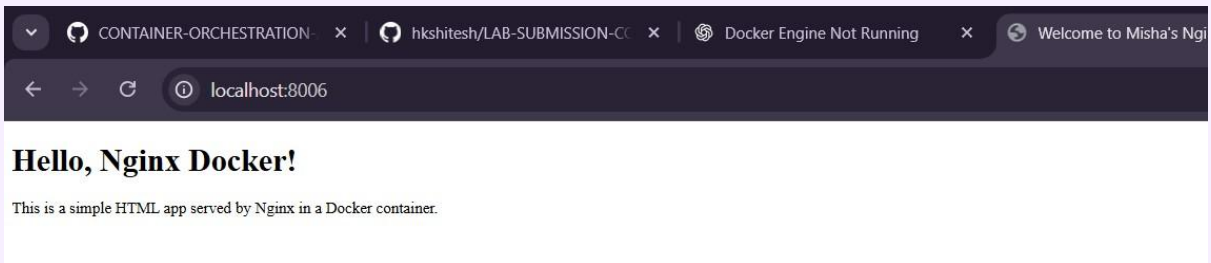
```
docker run -d -p 8006:80 nginx-html-app
```

```
PS D:\nginx-html-app-mi> docker run -d -p 8006:80 nginx-html-app-mi
c953bcfedcc9860738f455b6a4916f1f3e038ee82baca1fca1ccfec1ce9dd158
PS D:\nginx-html-app-mi>
```

This command runs the container in detached mode (-d) and maps port 8006 on your host machine to port 80 inside the container, where Nginx is serving your HTML app.

6. Step 5: Verify

Open a browser and go to http://localhost:8006. You should see your HTML page with the message “Hello, Nginx Docker!”.



7. Step 6: Stop and Remove the Container

Once you're done, you can stop and remove the container:

docker ps # to see running containers

```
PS D:\nginx-html-app-mi> docker ps  #
CONTAINER ID   IMAGE              COMMAND                  CREATED              STATUS              PORTS
                     NAMES
c953bcfedcc9   nginx-html-app-mi   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8006->80/tc
p, [::]:8006->80/tcp   pensive_jennings
PS D:\nginx-html-app-mi>
```

busybox	latest	2383baad1860	1 year ago	6.77 MB	
nginx-html-app-mi	latest	b184d0d81f70	3 minutes ago	236.8 MB	

Showing 12 items

RAM 1.35 GB CPU 0.08% Disk: 3.38 GB used (limit 1006.85 GB) Update available

docker stop <container-id>

docker rm <container-id>

```
p, [::]:8006->80/tcp   pensive_jennings
PS D:\nginx-html-app-mi> docker stop c953bcfedcc986
c953bcfedcc986
PS D:\nginx-html-app-mi> docker rm c953bcfedcc986
c953bcfedcc986
PS D:\nginx-html-app-mi>
```